Course-B.Tech	**Type-** Core
Course Code- CSET106	**Course Name-** Discrete Mathematical Structures
Year- 2026	**Semester-** Even
Date- 28/02/2026	**Batch-** 2025-2029

	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9
CO1									
CO2	✓	✓	✓	✓	✓	✓	✓	✓	✓
CO3			✓					✓	

Objectives

1. Students will be able to understand the integer arithmetic.
2. Students will be able to recognize the theory of congruences.
3. Students will be able to understand the applications related to congruence and inverse congruence.

Questions:

1. **Find the last two digits of the number**

$$9^{9^9}$$

2. **If a is an odd integer, then prove that** $a^2 \equiv 1 \ (mod\ 8)$.

3. **Perform the following operations in Zn:**
    a. **Add 9 to 8 in Z_{10}.**
    b. **85-28 (mod 20)**
    c. **91*89 (mod 30) (Note: Try to use congruence properties to simplify.)**

4. **Let n>1 be fixed and a, b, c, d be arbitrary integers then prove the following properties:**
    a. **If** $a \equiv b \ (mod\ n)$ **then**

    $a + c \equiv b + c \ (mod\ n)$ **and** $ac \equiv bc \ (mod\ n)$

    b. **If** $a \equiv b \ (mod\ n)$ **then for any positive integer k** $a^k \equiv b^k \ (mod\ n)$

5. **Using Euclidean algorithm find the GCD of the following:**
    a. **1475, 1200**
    b. **766, 1235**

6. **Use the Euclidean Algorithm to obtain integer x and y satisfying the following:**
    a. **GCD (119, 272) = 119 x + 272 y.**
    b. **GCD (1769, 2378) = 1769 x + 2378 y.**

7. **Solve the following linear congruence:**

   a. $18\,x \equiv \mathbf{30}\ (\boldsymbol{mod}\ \mathbf{42})$.

   b. $9\,x \equiv \mathbf{21}(\boldsymbol{mod}\ \mathbf{30})$.

8. **Find the multiplicative inverse of each of the elements of $\mathbf{Z_{12}}$.**

9. **In a multiplicative cipher, ciphertext is calculated from plaintext by using the relation C** $\equiv$ P*k (mod 26). What is the ciphertext for “India” by using key k=5? Here, alphabets are denoted as A = 1, B = 2, …, Y = 25, Z = 0.